

Factory Safety Incident Tracker

Week 1 Deliverable — Problem Definition and Scope

1. Problem Statement

Factories with manual or fragmented incident tracking (paper forms, spreadsheets, WhatsApp messages) suffer from delayed incident reporting, lost or duplicated records, and no real-time visibility for safety officers or plant managers. As a result, high-severity incidents can go unnoticed or unresolved for longer than acceptable, increasing risk to workers and compliance exposure. The Factory Safety Incident Tracker is a lightweight web application that lets shop-floor workers log safety incidents in real time, gives safety officers a searchable dashboard to triage and resolve them, and gives plant managers summary visibility into open risk — replacing ad-hoc, disconnected reporting with a single, auditable system.

2. Stakeholder List

Stakeholder	Role	Interest / Need
Shop-floor Worker	Incident Reporter	Fast, simple way to log an incident (under 2 minutes), no training required
Safety Officer	Reviewer / Resolver	Searchable dashboard, ability to update status, see high-severity items first
Plant Manager	Oversight	Summary indicators (open/closed/high-severity counts), drill-down for detail
IT / Admin	System Owner	Easy deployment, low maintenance, must run on existing Tomcat/Nginx infra
Project Team	Developer / DevOps	15-week feasible build with a fully demonstrable DevOps pipeline

3. Objectives (Measurable)

- 1 Reduce incident logging time to under 2 minutes per entry.
- 2 Achieve 100% searchability of incidents by date, status, severity, and location.
- 3 Dashboard and summary indicators load in under 3 seconds.
- 4 Zero data loss — every submitted incident is persisted and retrievable.
- 5 Deliver a fully automated DevOps pipeline (Git → Jenkins → Selenium → Docker → Ansible/Puppet) by Week 15, demonstrable live.

4. Constraints

- **Timeline:** Fixed 15 weeks — no scope creep past the frozen MVP.

- **Team size / skill:** Small team (or solo), learning Jenkins/Docker/Ansible alongside building — architecture must stay simple.
- **Budget:** No paid tools — must use free/open-source software (Jenkins OSS, Docker CE, Ansible, GitHub free tier).
- **Infrastructure:** Must be deployable to Tomcat or Nginx (per Week 3 requirement), later containerized.
- **No authentication / RBAC in MVP** — explicitly excluded to keep scope small (documented as a Week 15 future enhancement).
- **Single-node deployment** — no clustering or high-availability required for MVP.

5. Approved MVP Scope (Frozen)

Exactly five features are approved for the MVP. No additional features are to be added during the 15-week build; anything beyond this list is deferred to future enhancements.

- 1 **Incident / Event Entry** — form to log title, description, severity, location, and reporter.
- 2 **Searchable Dashboard** — filter and search incidents by date, status, severity, and location.
- 3 **Summary Indicators** — counts for open, closed, and high-severity incidents.
- 4 **Status Drill-Down** — clicking a summary indicator opens a filtered list, then a detail view.
- 5 **Alert / Exception View** — surfaces overdue or unresolved high-severity incidents.

Explicitly out of scope for MVP: mobile app, SMS/email notifications, user authentication/RBAC, multi-site support, analytics/reporting exports. These are documented as future enhancements, to be revisited in the Week 15 final report.